

Air Pollution

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Air pollution is contamination of the indoor or outdoor environment by any chemical, physical or biological agent that modifies the natural characteristics of the atmosphere. Household combustion devices, motor vehicles, industrial facilities and forest fires are common sources of air pollution. Pollutants of major public health concern include particulate matter, carbon monoxide, ozone, nitrogen dioxide and sulfur dioxide. Outdoor and indoor air pollution cause respiratory and other diseases and are important sources of morbidity and mortality.

Sources of Air Pollution

The sources of air pollution can be broadly classified into natural and anthropogenic. The natural sources include volcano, forest fire and pollens, organic compounds from plants, sea salt, suspended soils and dusts, etc. The anthropogenic sources include everything involving human activities such as:

Anthropogenic sources of air pollution:

1) **Burning of fossil fuels and motor vehicle exhausts:**

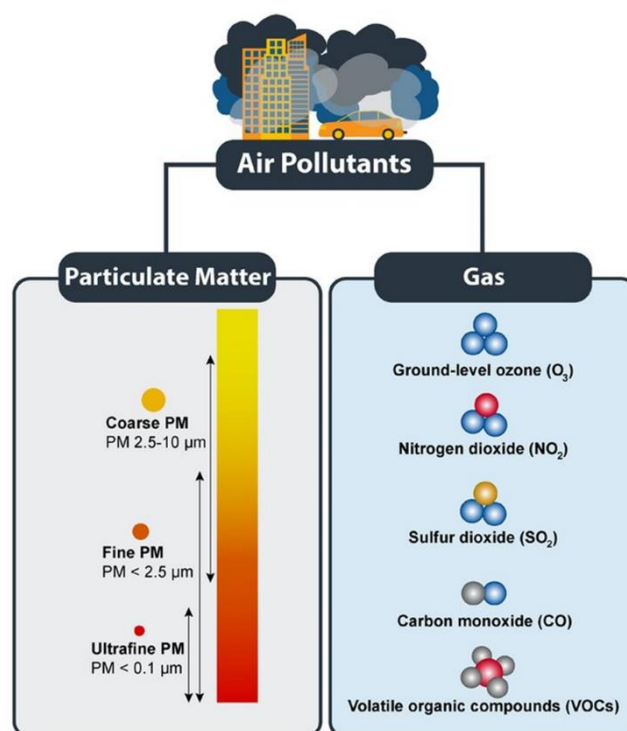
These can include gases emitted by the burning of fossil fuels and vehicle exhausts. This means that road traffic is one of the biggest sources of air pollution. Vehicles emit nitrogen oxides, carbon dioxide, carbon monoxide and particulate matter. Trains pollute a lot less than cars. But they also cause pollution, since they utilize a large amount of electricity produced by power stations, which produces large quantities of nitrogen oxides, carbon dioxides, sulfur dioxides and particulate matter. These pollutants may be categorised as primary or secondary pollutants.

2) **Agriculture:**

Animals like cows and sheep release a massive amount of methane through belching and breaking wind. Methane is a colourless gas which is produced in their stomachs when bacteria break down the food that they eat. Across the whole world, livestock is the biggest source of methane. Methane is the second most important greenhouse gas which can cause climate change.

3) **Waste disposal:**

Waste disposal from landfills is the largest producer of methane emitted after agriculture and livestock rearing.



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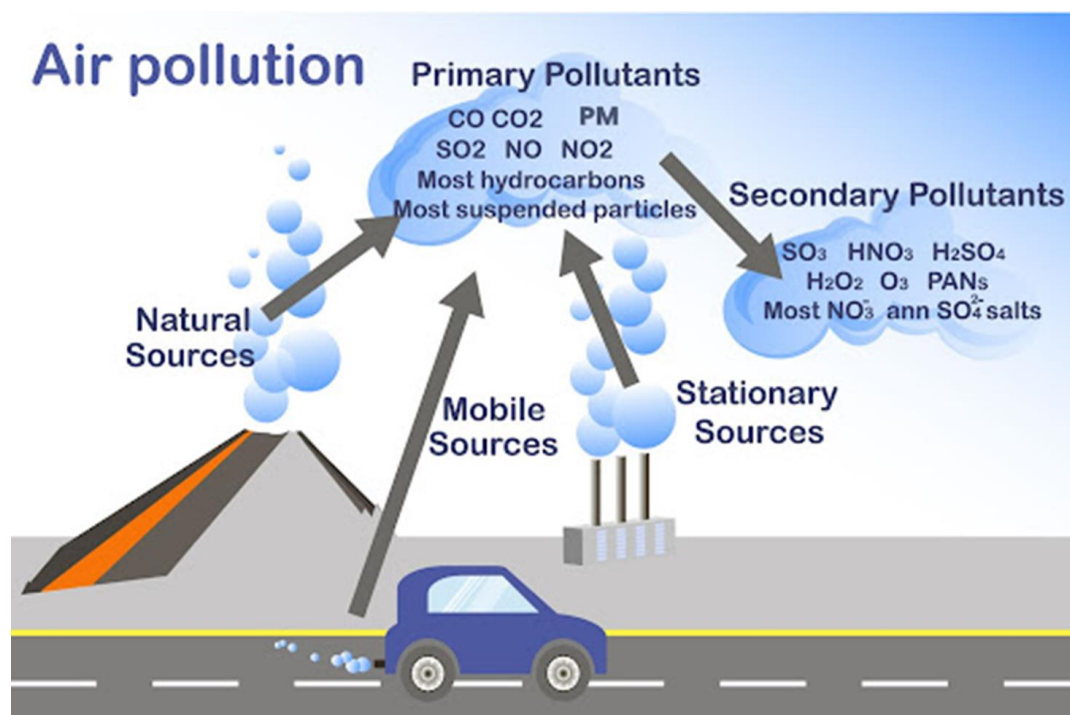
Primary Pollutants

Primary pollutants are directly emitted to the atmosphere. Air pollutants may have a natural, anthropogenic or mixed origin, depending on their sources or the sources of their precursors. Key primary air pollutants include particulate matter (PM), black carbon (BC), sulphur oxides (SO_2), nitrogen oxides (NO_x) (including nitrogen monoxide and nitrogen dioxide, NO_2), ammonia (NH_3), carbon monoxide (CO), methane (CH_4), non-methane volatile organic compounds (NMVOCs), including benzene, and certain metals and polycyclic aromatic hydrocarbons, including benzo[a]pyrenes (BaP).

Secondary Pollutants

Secondary pollutants are formed in the atmosphere from precursor gases through chemical reactions and microphysical processes. Key secondary air pollutants are PM, ozone (O_3), NO_2 and several oxidised volatile organic compounds (VOCs). Key precursor gases for secondary PM are sulphur dioxide (SO_2), NO_x , NH_3 and VOCs. These pollutants and their precursor gases can be of both natural and anthropogenic origin including:

- Burning of fossil fuels in electricity generation, transport, industry and households
- Industrial processes and solvent use, for example in the chemical and mining industries;
- Agriculture
- Waste treatment
- Natural sources, including volcanic eruptions, windblown dust, sea-salt spray and emissions of volatile organic compounds from plants



Primary and Secondary Air Pollutants

Reprinted from: <https://www.clarity.io/blog/how-measuring-different-types-of-air-pollutants-creates-a-more-holistic-picture-of-air-pollution>

Types of Air Pollutants

The following pollutants form the major category of air pollutants.

Carbon monoxide (CO):

Carbon monoxide (CO) is a colorless, odorless, and highly toxic gas that forms during the incomplete combustion of carbon-containing materials. Major sources are motor vehicle exhaust, burning of forests and grasslands, smokestacks of fossil fuel-burning power plants and industries, tobacco smoke, and open fires and inefficient stoves used for cooking. Carbon monoxide can combine with hemoglobin in red blood cells, which prevents the normal binding of oxygen with hemoglobin molecules. This in turn reduces the ability of blood to transport oxygen to body cells and tissues. Long-term exposure can trigger heart attacks and aggravate lung diseases such as asthma and emphysema. At high levels, CO can cause headache, nausea, drowsiness, confusion, collapse, coma, and death.

Carbon dioxide:

Carbon dioxide (CO₂) is a colorless, odorless gas. About 93% of the CO₂ in the atmosphere is the result of the natural carbon cycle. The rest comes from human activities, mostly the burning of fossil fuels and the clearing of CO₂-absorbing forests and grasslands. CO₂ is being added to the atmosphere faster than it is removed by the natural carbon cycle. This can contribute to human health problems such as heat exhaustion and to the reduction of food supplies in some areas, while causing water shortages, prolonged drought, or excessive flooding in other areas.

Nitrogen oxides:

Nitrogen oxides are emitted during fuel combustion from industrial facilities and the road transport sector. NO_x is a group of gases comprising nitrogen monoxide (NO) and nitrogen dioxide (NO₂). NO makes up the majority of NO_x emissions. NO_x contributes to the formation of ozone and particulate matter.

Sulfur dioxide:

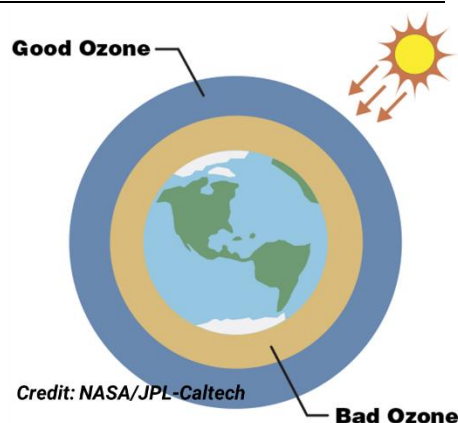
Sulphur dioxide is formed and emitted by combustion of fossil fuels (mainly coal and oil) primarily for electricity generation. High concentrations of SO₂ are associated with multiple health and environmental effects. The highest concentrations of SO₂ have been recorded in the vicinity of large industrial facilities. SO₂ emissions are an important environmental issue because they are a major precursor to ambient PM_{2.5}.

Ground-level ozone:

Ground level ozone is created when sunlight reacts with, volatile organic compounds (VOCs) and nitrous oxides (NO_x). When particles in the air combine with ozone, they create smog. Smog is a type of air pollution that looks like smoky fog and makes it difficult to see. These can be transported long distances by wind.

Photochemical smog:

Photochemical smog also results from interactions between different air pollutants. This smog has a brown haze and can be painful to the eyes, accounting for most of the smog we see today. Photochemical smog forms from interactions between particulates, nitrogen oxides, ozone, and other air pollutants, though primarily from VOCs and NO_x since ozone comprises a large portion of this smog.





Beijing on a clear day after a rain (left) and on a smoggy day (right)

Credit: [Bobak](#) via Wikimedia Commons

Suspended Particulate Matter (SPM):

Particulate matter is a mixture of aerosol particles (solid and liquid) covering a wide range of sizes and chemical compositions. PM is either directly emitted as primary particles or it forms in the atmosphere from emissions of certain precursor pollutants such as SO₂, NO_x, NH₃. SPM is emitted from many anthropogenic sources, including both combustion and non-combustion sources. Natural emissions of PM also occur, including from sea salt and windblown Saharan dust.

Volatile Organic Compounds (VOCs)

Volatile organic compounds (VOCs) are emitted as gases from certain solids or liquids. VOCs include a variety of chemicals, some of which may have short- and long-term adverse health effects.

Management of Air Pollution

Air pollution management aims at the elimination or reduction to acceptable levels, of airborne gaseous pollutants, suspended particulate matter and physical and, to a certain extent, biological agents whose presence in the atmosphere can cause adverse effects on human health, deleterious effects on animal or plant life, damage to materials of economic value to society and damage to the environment

Health Effects of Air Pollution

Our body has a number of natural defence mechanisms to help protect us against air pollution. But prolonged or acute exposure to air pollutants, including tobacco smoke can overload or break down these natural defenses.

- Years of smoking or breathing polluted air can lead to other lung ailments such as chronic bronchitis and emphysema, which leads to acute shortness of breath and usually to death.
- Inhalation of small, fine and ultra-fine particles added to the atmosphere by coal-burning power plants causes asthma attacks and other respiratory disorders.

Steps to Reduce Air Pollution

The best air quality management methods stress that the air pollutant emissions should be kept to a minimum. Some of the methods that can be used to reduce or minimize air pollution are:

- Enforcement of the use of catalytic converters in vehicles or of emission standards in incinerators
- Shut-down of factories or reduction of traffic during unfavourable weather conditions
- Strict laws for emission of pollutants, which emphasize prevention of emission
- Stricter laws need to be enforced on coal-burning power plants and industrial facilities so that the harmful emissions of sulfur dioxides and nitrogen oxides can be controlled
- Use of air pollution control devices such as chemical scrubbers in emission towers that can capture most of the harmful chemicals that might be emitted in industries. E.g., SO_2 can be removed by use of a lime scrubber
- Control devices such as inertial separators for particular matter and wet collectors
- Safe disposal methods to reduce the effects of the harmful agents
- Tax each unit of pollutant produced

We, as individuals can take a few steps to reduce consumption of energy and air pollution. They are summarized as follows.

1. Walk, bike or use public transportation to reduce air pollution
2. Minimize pollution from cars by prevention of idling
3. Save energy and make sure you use energy efficiently
4. Recycle and reuse
5. Consume less and choose sustainable products
6. Avoid/minimize plastic bags
7. Reduction of forest fires and smoking
8. Use of fans instead of Air conditioners
9. Use filters for chimneys
10. Avoid usage of crackers
11. Avoid using of products with chemicals
12. Implement Afforestation